Session 2018-2019.

4. a)

I. What are the most important skills that a system analyst must have?

- **Technical knowledge:** A system analyst should have a strong understanding of computer systems, software development methodologies, and programming languages.
- **Analytical skills:** System analysts need to be highly analytical and detail-oriented. They must be able to analyze complex problems, break them down into smaller components, and identify patterns and relationships.
- **Business acumen:** Understanding the business context is crucial for a system analyst. They should possess a good understanding of the organization's goals, processes, and industry requirements.
- **Communication and interpersonal skills:** System analysts often interact with various stakeholders, including clients, users, developers, and management. Effective communication skills are vital for gathering requirements, explaining technical concepts to non-technical individuals, and collaborating with different teams.
- **Problem-solving and critical thinking:** System analysts encounter complex problems that require innovative and logical solutions. They need to think critically, analyze options, and evaluate the potential impact of their decisions.
- **Project management:** System analysts are often involved in projects that require managing timelines, resources, and deliverables.
- **Adaptability and learning agility:** Technology and business requirements evolve rapidly. System analysts need to stay updated with the latest industry trends, emerging technologies, and new methodologies.

II. Describe different types of questionnaires. What are the advantages and disadvantages of questionnaires?

- **Free-format questionnaires** offer the respondent greater latitude in the answer. A question is asked, and the respondent records the answer in the space provided after the question.
  - **Types of free-format questions:**
    - ✓ Opinion question.
    - ✓ Exploratory question.
    - ✓ Scenario-based question.
    - ✓ Descriptive questions.
    - ✓ Comparative questions.

- **Fixed-format questionnaires** contain questions that require selection of predefined responses from individuals.
    - **Types of fixed format questions:**
        - ✓ Multiple-choice questions
        - ✓ Rating questions
        - ✓ Ranking questions

➢ Advantage.

- Can be answered quickly.
- Inexpensive means.
- Real facts expressed.
- Responses can be tabulated and analyzed quickly.

➢ Disadvantage.

- Respondents often low
- No guarantee to get answer or expand all question
- No opportunity for voluntary information
- Not possible to read body language
- Good questions are difficult to prepare.

b)

I. What is present value of money? How we can calculate present value of money?

The present value of money, also known as the present discounted value, is a financial concept that refers to the value of a future sum of money in terms of its worth in today's dollars. It takes into account the time value of money, which means that money available in the present is generally more valuable than the same amount of money in the future due to factors like inflation and the opportunity cost of investing or earning interest.

To calculate the present value of money, you need to know the future value, the interest rate or discount rate, and the time period involved. The most commonly used formula to calculate present value is the discounted cash flow (DCF) formula:

PV = FV / (1 + r)^n

Where:

- ✓ PV = Present value
- ✓ FV = Future value (the amount of money you want to calculate the present value for)
- ✓ r = Interest rate or discount rate (expressed as a decimal)
- ✓ n = Number of periods (usually in years)

The formula discounts the future value by dividing it by the factor (1 + r)^n, where (1 + r) represents the present value of money and ^n represents the number of periods. By discounting the future value, the formula takes into account the time value of money and calculates its present value.

II. What should do and avoid for an interview session?

- **Interviewing Do's and Don'ts:**

Do	Avoid
• Be courteous (polite) • Listen carefully • Maintain control • Probe (inquery) • Observe mannerisms and nonverbal communication • Be patient • Keep interviewee at ease • Maintain self-control	• Continuing an interview unnecessarily. • Assuming an answer is finished or leading nowhere. • Revealing (express) verbal and nonverbal clues. • Revealing your personal biases. • Talking instead of listening. • Assuming anything about the topic and the interviewee. • Tape recording -- a sign of poor listening skills.

5. a)

I. "A dollar today is worth more than a dollar one year from now"-what is the significance of this statement?

Let's take an example to illustrate this. Suppose we have $1,000 that will be received three years from now, and the discount rate is 5%. Using the formula above:

We know that, PV = FV / (1 + r)^n

Where:

- ✓ PV = Present value
- ✓ FV = Future value (the amount of money you want to calculate the present value for)
- ✓ r = Interest rate or discount rate (expressed as a decimal)
- ✓ n = Number of periods (usually in years)

PV = $1,000 / (1 + 0.05)^3
PV = $1,000 / (1.05)^3

PV = $1,000 / 1.157625
PV ≈ $863.84

So, the present value of $1,000 to be received three years from now, with a discount rate of 5%, is approximately $863.84.

The formula discounts the future value by dividing it by the factor (1 + r)^n, where (1 + r) represents the present value of money and ^n represents the number of periods. By discounting the future value, the formula takes into account the time value of money and calculates its present value.

II. What are the advantages and disadvantages of interviews?

Interviews are a fact-finding technique whereby the systems analysts collect information from individuals through face-to-face interaction.

- **Advantages of interviews:**
  - ✓ Opportunity to motivate the interviewee
  - ✓ More feedback from the interviewee
  - ✓ Permit system analyst to adapt or reward questions for each individual
  - ✓ possible to read body language
- **Disadvantages of interviews:**
  - ✓ Time consuming
  - ✓ Success highly depends on system analysts' human relation skill
  - ✓ May be impractical due to location.

b) Define the following:

I. **Proxemics:** Proxemics is the study of how space is used in human interactions. For example, authority can be communicated by the height from which one person interacts with another. If one stands while the other sits or lies down, the person standing has placed himself or herself in a position of authority.

II. **Brainstorming:** Brainstorming is a technique for generating ideas during group meetings.

III. **Body language:** Form of nonverbal communication that
- ✓ We all use & are usually unaware of
- ✓ By research
  - Verbally-7 % (in word)
  - Tone of voice- 38 %
  - Facial & Body expressions- 55%

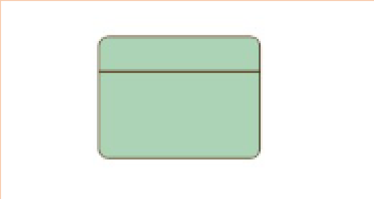
6. a)

I. How many symbols are used for DFD? Make comparison between DFDs and ERD?

Three symbols and one connection:

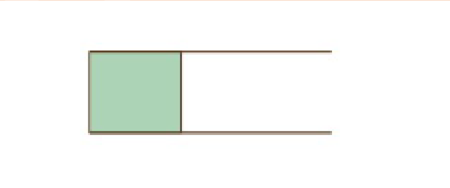
- **External entities** are represented by squares as the source or destination of data.

- **Processes** are represented by rectangles with rounded corners.



- **Data Flows** are referred to by arrows to denote the physical or electronic flow of data.

- **Data Stores** are physical or electronic-like XML files denoted by open-ended rectangles.



➢ **DFD and ERD comparisons.**

S.No.	DFD	ERD
1.	It stands for Data Flow Diagram.	It stands for Entity Relationship Diagram or Model.
2.	Main objective is to represent the processes and data flow between them.	Main objective is to represent the data object or entity and relationship between them.

3.	It explains the flow and process of data input, data output, and storing data.	It explains and represent the relationship between entities stored in a database.
4.	Rule followed by DFD is that at least one data flow should be there entering into and leaving the process or store.	Rule followed by ERD is that all entities must represent the set of similar things.
5.	It models the flow of data through a system.	It model entities like people, objects, places and events for which data is stored in a system.

II. Differentiate between databases and conventional files.

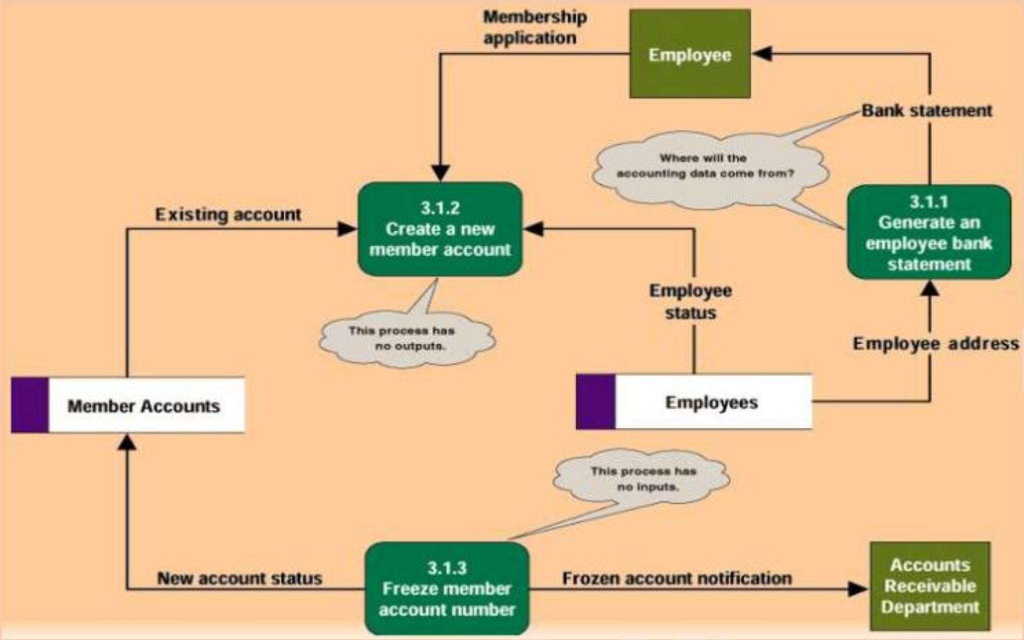
Basis	File System	DBMS
Data Redundancy	Redundant data can be present in a file system.	In DBMS there is no redundant data.
Query processing	There is no efficient query processing in the file system.	Efficient query processing is there in DBMS.
Cost	It is less expensive than DBMS.	It has a comparatively higher cost than a file system.
Data Independence	There is no data independence.	In DBMS data independence exists.
Example	Cobol, C++	Oracle, SQL Server

b)

I. What are the most common process occur when a DFD are drawn for a system?

1. **Black hole:** 3.1.2 has inputs but no outputs (it is called black hole because data enter the process and then disappear).
2. **David copperfield:**3.1.3 has outputs but no inputs. Unless you are David Copperfield (most commercially successful magician in history) it's a miracle.

3. **Gray hole:** 3.1.1 the inputs are insufficient to produce the o/p (it is called gray hole because
    - ✓ Misnamed process
    - ✓ Misnamed inputs and/or outputs
    - ✓ Incomplete facts



II. "The time value of money is not taken into account for Payback Analysis"-Explain the statement with appropriate example.

The statement "The time value of money is not taken into account for Payback Analysis" means that the Payback Analysis method does not consider the concept of the time value of money when evaluating investment projects.

Let's understand this with an example:

Suppose you are considering two investment projects, Project A and Project B, both requiring an initial investment of $10,000. Project A is expected to generate equal annual cash flows of $2,000 for five years, while Project B is expected to generate equal annual cash flows of $1,500 for eight years.

Using Payback Analysis, we calculate the payback period for each project by determining how long it takes to recover the initial investment. In this case, both projects have a payback period of 5 years since it takes five years for the cumulative cash inflows to equal or exceed the initial investment of $10,000.

However, Payback Analysis does not consider the time value of money. It treats all cash inflows equally, regardless of when they occur. This means that the method does not account for the fact that money received earlier is generally more valuable than money received later due to factors like inflation, the opportunity cost of capital, or the potential to invest the money elsewhere.

To illustrate this, let's consider the time value of money by discounting the cash flows at a 10% discount rate. The present value (PV) of each cash flow can be calculated as follows:

$PV = CF / (1 + r)^n$

Using the formula, we find the present value of the cash flows for both projects:

For Project A:
$PV = \$2,000 / (1 + 0.10)^1 + \$2,000 / (1 + 0.10)^2 + \$2,000 / (1 + 0.10)^3 + \$2,000 / (1 + 0.10)^4 + \$2,000 / (1 + 0.10)^5$
$PV \approx \$7,513.19$
For Project B:
$PV = \$1,500 / (1 + 0.10)^1 + \$1,500 / (1 + 0.10)^2 + ... + \$1,500 / (1 + 0.10)^8$
$PV \approx \$9,042.11$

From the present value calculations, we can see that Project A has a lower present value ($7,513.19) compared to Project B ($9,042.11). Considering the time value of money, Project B has a higher present value, indicating that it generates more value in today's dollars compared to Project A.

Session 2017-2018.

4 A.

a) In system analysis if you use questionnaire as fact finding process which advantages and disadvantages you can get?

Already noted

b) How reports differ for different persons and places?

4 B.

I. What are the cases where illegal data flows happen?

Already noted

II. **Explain the following terms with appropriate example:**
   a) **Body language.**
   b) **Spatial zones.(from shortlist pdf note)**

Already noted

5 A.

I. **Show with example what are the most common process errors occur when a data flow Diagram are drawn for a system.**

Already noted

II. **Define the following:**
   a) **Proxemics.**
   b) **Brainstorming.**

Already noted

5 B.

I. **How DFD differ from ERD?**

Already noted

II. **What are the advantages and disadvantages of interviews over the interview processes?**

Already noted

6 A.

i. **"A dollar today is worth more than a dollar one year from now"-what is the significance of this statement?**

Already noted

ii. **What are the significances of database integrity in system analysis and design?**
   - **Data Accuracy:** Database integrity helps maintain the accuracy of data by enforcing constraints and rules on the data stored in the database. It ensures that data is entered correctly, follows specified formats, and meets predefined criteria.
   - **Data Consistency:** Database integrity ensures data consistency by enforcing relationships and dependencies between different data entities. It prevents data

anomalies, such as duplicate records, conflicting information, or data inconsistencies that can arise from data redundancy or data manipulation.

- **Data Reliability:** Database integrity ensures data reliability by protecting against data corruption or unauthorized modifications. It includes mechanisms such as data validation, access controls, and data encryption to ensure that only authorized users can access and modify the data.
- **Data Security:** Database integrity plays a significant role in data security. By enforcing data integrity rules, it helps prevent unauthorized access, data breaches, or data loss.
- **Data Completeness:** Database integrity ensures data completeness by enforcing constraints and validations to prevent missing or incomplete data.
- **Data Auditability:** Database integrity enables the tracking and auditing of data changes, providing a record of who made changes, when they were made, and what changes were made.

6 B.

i. **"The time value of money is not taken into account for Payback Analysis"-Explain the statement with appropriate example.**

ii. **Differentiate between databases and conventional files.**

Already noted